# Fiber Optics - Modal dispersion report

Diego A. Diaz Riega, Francesco Lorenzi, Francesco Nardo and Pietro Metuh

## I. Introduction

Multi-mode fibers are typically characterized by a sufficiently larger core diameter than single-mode fibers, in order to support other propagating modes. However, the maximum length of a transmission link in multi-mode fibers is severely reduced because of modal dispersion; in particular, the product between the bandwidth and the maximum length of multi-mode fibers decreases, limiting the channel capacity.

Modal dispersion manifests itself by spreading the signal in time, due to the propagation constant of the optical signal not being equal for all modes. It should not be confused with chromatic dispersion, which is caused by the frequency dependence of the group velocity itself, and is typically a secondary effect with respect to modal dispersion. By contrast, modal dispersion occurs even with an ideal, monochromatic light source.

A special case of modal dispersion is polarization mode dispersion (PMD), which is present even in single-mode fibers. PMD results when two ideally-degenerate modes that are thus normally characterized by the same propagation constant actually become non-degenerate, possibly due to asymmetries in the fiber, stress, bends and many other causes that can break the cylindrical symmetry of the fiber axis.

## II. Experimental setup

In this experiment, we considered two different multi-mode fiber bobbins, which will be identified by the color of their coil. The specifications of these fibers are shown in table I (the estimated bandwidth values are obtained from the analysis in the following section).

| Fiber | Blue | Red |
|---|---|---|
| Length [km] | 4.244 | 2.517 |
| Profile Type | Graded-index | Step-index |
| Bandwidth | 670 MHz-km | 221 MHz-km |

TABLE I: Parameters of the two fiber bobbins used during this experience.

This laboratory experience is divided into two main parts:

1) Measurement and analysis of the amplitude (i.e. the real part) of the spectral response of the multi-mode fibers and characterization of the fibers by their 3-dB bandwidth, expressed in [MHz-Km].
2) Measurement of the noise caused by modal dispersion, by analyzing the bit-error rate (BER) of a given transmission.

For the first part, we employed an electrical spectrum analyzer (ESA), which pilots the laser through a radio frequency modulator. The modulating frequency of this device varies in the range $[0.01 \div 2]$ GHz; as a matter of fact, obtaining a lower modulating frequency is nontrivial, and a zero-Hertz minimum in modulating frequency cannot be achieved by a physical device. After the conversion of light back to an electric signal performed by the photodiode, the ESA reads the incoming data and plots the real part of the amplitude of the spectral response. Both the fibers were tested with the same laser diode, which emits at a fixed wavelength of 1330 nm, and the same photo-diode.

The second part of this experience employed a slightly different setup: instead of an ESA, a BER tester was used to test and compare the performance of a link made with the red and the blue fibers.

Unfortunately, because of some issues with the experimental setup during this experience, we were not able to observe and measure our own data. The traces were thus given to us after the experience. Nevertheless, this part of the experiment mainly deals with post-processing and data analysis. As a matter of fact, by comparing the original and the post-transmission traces of a signal we were able to determine the so-called *eye diagram*, the *quality factor* $Q$ and finally the *bit-error rate* (BER) (more details will be provided in Section III). A picture of the setup that has just been described is shown in Fig.1.

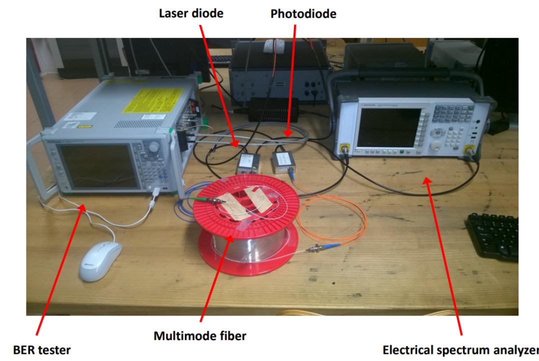


Fig. 1: Experimental setup.

## III. Results

### A. Spectral Response

As previously stated, the first part of the experiment deals with the comparison of spectral responses between the blue and the red fiber. In order to do this, we first obtained the data of the spectral responses from the ESA given by the blue fiber. The same procedure was repeated with the red fiber.

Moreover, a back-to-back (B2B) fiber, namely a fiber that is merely a few decimeters long, was used to simulate a direct connection between the laser and the photodiode. The purpose of this fiber is to measure the noise caused by the other related devices, such as the laser and the photodiode, so that it can be accounted for in the analysis. By subtracting the spectral

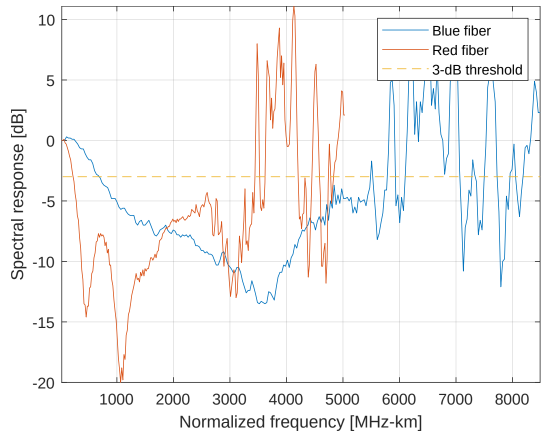


Fig. 2: Comparison of the spectral responses.

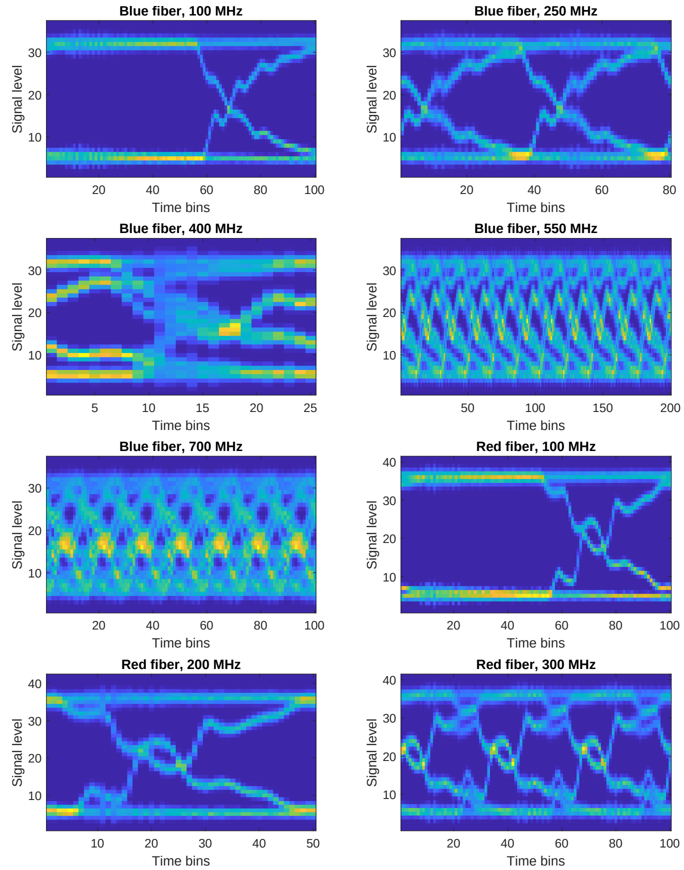


Fig. 3: Eye diagrams from the data samples.

response of this last scenario to the other two, the spectral responses of the two fibers are obtained.

The last post-processing step is to normalize the frequency by multiplying it by the length of the fiber. As a matter of fact, the bandwidth parameter of commercially-available fibers is usually given in MHz-km, since the available bandwidth decreases with the length of the fiber.

After these steps, the amplitude of the response as a function of normalized frequency is obtained, as shown in Figure 2. As expected, the performance of the red fiber is considerably poorer than the blue fiber, because in general step-index fibers suffer from modal dispersion more than graded-index fibers.

In particular, the cut-off frequency of the blue graded fiber (670 MHz-km) is more than three times larger than that of the red step-index fiber (221 MHz-km). Moreover, the obtained values are also sensible with the average 400 MHz-km bandwidth typically found in step-index fibers.

Finally, by looking at the same figure, we can observe that there is also a noticeable increase of noise with the frequency for both fibers, once the minimum has been reached.

### *B. BER tester*

As mentioned at the end of Section II, we experienced technical issues during the lab experience, but a complete analysis of the traces (of both the original electric signal and the signal after propagation along the fiber) that were provided to us, was still possible.

Overall, eight traces are considered. Five of them are the time signals measured by the BER tester and were taken after applying a modulating signal with a frequency of 100, 250, 400, 550 and 700 MHz. The other three correspond to the red fiber, which was tested at 100, 200 and 300 MHz.

The first step of the post-processing was to build the eye diagrams for the traces, namely to plot the time signal over the same interval for several periods, after normalizing the traces by subtracting the minimum. The overall number of time bins corresponds to at least one period, and is measured in multiples of the sampling period $T_s = 100$ ps.

Since not all the modulating frequencies are multiples of the sampling frequency $f_s = 10$ GHz, the number of bins is further extended in order not to make the signal out of phase. For instance, for each bit period in the at 550 MHz, $18.\overline{18}$ samples are measured; in other words, every 5.5 bits, an additional integer sample is recorded. Hence, 11 bits of 18 samples are represented, plus two additional ones to restore stability in the diagram.

All the eye diagrams are shown in figure 3, with $10^7$ samples analyzed for each trace. Naturally, the higher the modulating frequency, the narrower the eye, meaning that the distinction between a zero and a one is less sharp.

The eye diagram is a great tool in telecommunications, and from it several performance phenomena can be determined. However, a more objective measure of the signal quality is given by the BER, which can be calculated via further manipulation of the diagram.

Ideally, for a given time bin of the eye diagram, the distribution of the signal level represents two Gaussian probability density functions $\mathcal{N}(I_0, \sigma_0^2)$ and $\mathcal{N}(I_1, \sigma_1^2)$, each representing the probability of deciding for a zero or a one, respectively. If the decision threshold that minimizes the BER is chosen, then this parameter is given by

$$\mathrm{BER} = \frac{1}{2}\mathrm{erfc}\left(\frac{Q}{\sqrt{2}}\right), \tag{1}$$

where erfc is the complementary error function and $Q$ is the quality factor, defined as

$$Q = \frac{I_1 - I_0}{\sigma_1 - \sigma_0}. \tag{2}$$

| Fiber type | Modulating frequency | Maximum Q-factor | BER |
|---|---|---|---|
| Blue | 100 MHz | 18.30 | 4.3e-75 |
| Blue | 250 MHz | 12.20 | 1.6e-34 |
| Blue | 400 MHz | 4.23 | 1.1e-05 |
| Blue | 550 MHz | 2.67 | 3.7e-03 |
| Blue | 700 MHz | 2.27 | 1.2e-02 |
| Red | 100 MHz | 18.21 | 2.2e-74 |
| Red | 200 MHz | 13.90 | 3.2e-44 |
| Red | 300 MHz | 4.56 | 2.6e-06 |

TABLE II: Quality parameters from each dataset.

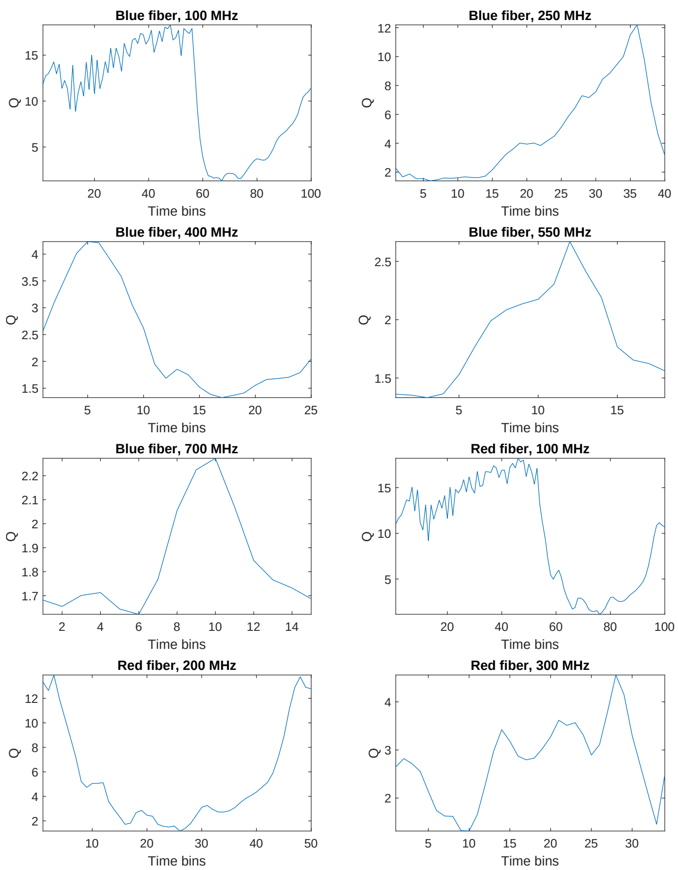


Fig. 4: Quality factors as a function of the position along the diagram.

Two Gaussian fits were thus performed to obtain mean, variance and finally the quality factor, for each bin of a single eye-diagram period. By varying the threshold value $I_D$ over an interval and accepting the maximum quality factor, we optimized the decision regions and obtained the largest Q value. Compared with the results obtained from a simpler decision rule of $I_D = (I_{min} + I_{max})/2$, the quality factor is slightly increased for the blue fiber at 550 and 700 MHz, but the improvement is marginal.

The quality factors are plotted in Figure 4. In general, the data taken with a lower modulation frequency are not only characterized by a higher Q value, but they are also great over a wider portion of the time bin interval. By contrast, higher modulation frequencies are characterized by sharp maxima, because of the narrower distribution which is also present in the eye diagrams.

Finally, the BER is calculated by picking the highest quality factor; the resulting values are shown in Table II, and a visual plot is picture in Figure 5. Even though the number of data points is not very high to make a systematic comparison, it can still be clearly observed that the performance of the red fiber is consistently poorer than the blue one, as determined in the first part of the analysis.

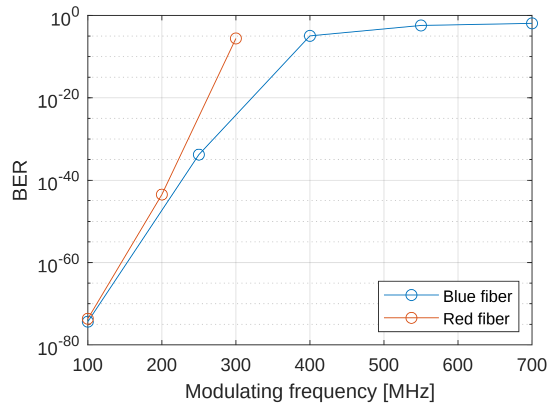


Fig. 5: BER as a function of the modulating frequency.

## IV. Conclusion

In this experiment we evaluated the performance of two different multi-mode fiber (specifications in Table I). We considered the spectral response and the bit-error ratio from both of them, thus underlining the effect of modal dispersion.

First, from the spectral analysis we discovered that the blue fiber has a wider bandwidth (670 MHz-km) compared to the red one (221 MHz-km), as expected by their respective characteristics (graded index vs step index).

Secondly, after building the eye-diagram we ended up calculating the Q-factor, which is not constant but indeed related to the modulating frequency. Then, by considering the maximum Q-factor for each modulating frequency we evaluated the corresponding BER by applying (1). The results (summarized in Table II) confirm that overall the blue bobbin performs better than the red one at a given modulating frequency, and the gap between the two increases with the modulating frequency.

The values of the maximum acceptable BER for a stable channel usually depend on the modulation scheme. For instance, if a simple on-off keying (OOK) scheme were to be applied, giving the typical requirement of a BER of about $10^{-6}$, the red fiber could be used with a modulating frequency of up to 300 MHz, while the blue fiber could withstand modulation at up to about 390 MHz.

In conclusion, a multi-mode fiber supports more than one propagation mode and potentially multiple communication channels, given the orthogonality of the modes. However, the maximum length of the transmission link is limited due to modal dispersion which affects the fiber bandwidth. Furthermore, as seen in the second part of the analysis, a too-high modulating frequency makes the communication channel unstable because of the increased probability of wrongly reading the value of a bit, thus limiting the maximum theoretical transmission efficiency.